

A Novel Approach for Classification of Hand Movements using Surface EMG Signals

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Abstract—Surface Electromyography (EMG) signals are being used in various fields including myoelectric control for prosthetic devices, human machine-interaction and other clinical and biomedical applications. EMG signals acquired through surface electrodes include various sources of noise like electrode-skin interface and electromagnetic noise etc. Therefore, accurate processing and classification of EMG signals is a challenging step in order to achieve reliable results for myoelectric control and so on. This paper introduces a novel classifier using Stacked sparse autoencoders (SSAEs) for improved myoelectric control. Data of six surface EMG channels is collected from five able-bodied and two amputee subjects for two days using their right hand or the amputated one. For performance evaluation, offline classification error is used and performance of SSAEs is compared with state of the art LDA. For within day analysis, SSAEs ($1.29 \pm 0.83\%$) outperformed LDA ($4.09 \pm 2.15\%$) with p value of 0.0018 for both able-bodied and amputee subjects. In between days' analysis, SSAEs outperformed ($P < 0.001$) LDA for both able-bodied and amputee subjects. These findings suggest that deep features of Autoencoders have the potential to be used as control source for myoelectric control systems and it can significantly improve the performance as compared to classical machine learning algorithm.

Keywords—Electromyography, upper limb prosthetics, Stacked Sparse Autoencoders, myoelectric control, deep features learning.

I. INTRODUCTION

Electromyographic signals are the electrical signals that are generated due to contraction of muscles which are controlled by nervous system [1]. These signals are being used for control of robotic hands since decades. They are recorded using either surface or needle electrodes, also called surface electromyography (sEMG) and intramuscular (imEMG) respectively. sEMG is the non-invasive [2] source for myoelectric control of upper limb prosthetics and is being more commonly used in the literature on myoelectric control [3]. On the other hand, imEMG is an invasive technique [4] and provide certain benefits over sEMG like little crosstalk and stable control signals from deep muscles [4-6]. But due to its limited information from specific sites, it may also be a disadvantage

unlike sEMG which include more information from muscles [7].

Different myoelectric control schemes have been introduced that utilize either surface or intramuscular EMG signals as control source [8, 9]. A control system for robotic hand may be either sequential or simultaneous in nature [8]. Most of commercially available schemes adopt simple strategy to encode the EMG signals recorded from one or more sites [10], but they are not robust enough to accurately control the more functionality of dexterous prosthetic hands [11, 12]. Pattern recognition based control strategies [13-16] have shown promising results with the sEMG based control schemes and certain studies [3, 10, 17-20] showed that either sEMG performed better than imEMG or there was no significant difference found between them. In such schemes, features are extracted from raw EMG signals and hence they provide comparatively more and specific information to utilize greater functionality of prosthetic hand.

An accurate classification of hand movements is very crucial because a misclassification of movement can be very dangerous for the robotic hand's user [21]. Different classifiers including Artificial neural network (ANN), Linear discriminant analysis (ANN), support vector machine (SVM), self-organizing map (SOP) and fuzzy classifiers are being used for signals classification [1]. Chen and Green [22] developed a state of art LDA classifier in order to provide benchmark against other developing algorithms. Autoencoders are being successfully used in Electroencephalography [23, 24], Electrocardiography [25] and in other fields of biomedical signals processing [26-28] but it is less utilized in EMG.

In this work, two layers stacked sparse autoencoders (SSAEs) has been developed in order to explore its application for improved myoelectric control and its performance is compared with LDA. For both the classifiers, sEMG data of five abled and two amputee subjects is used as control source. In sections 2, methodology of the work is explained and results are presented in section 3.

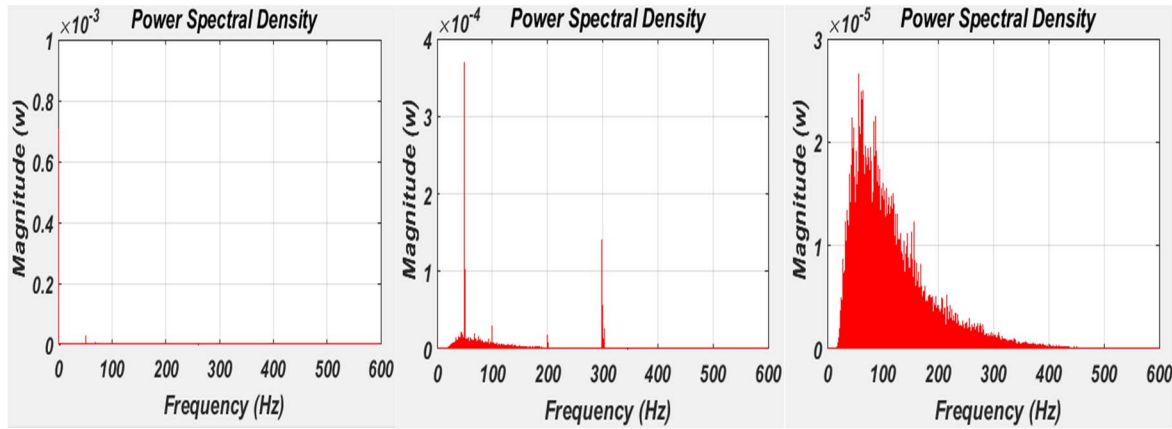


Fig. 1. It shows the application of configurable Butterworth filter to raw EMG data. left figure shows the raw EMG signal with motion artifacts. Central figure shows the result after bandpass filtering while right figure shows the EMG signal after noise removal with configurable Butterworth bandstop filter.

II. METHODOLOGY

A. Data collection and Experimental procedure

Five able-bodied and two transradial amputee subjects participated in this study. For experimentation, right hand of all subjects were used except for one amputee subjects where his left hand was used. The procedures were in accordance with the Declaration of Helsinki and were approved by the ethical committee of Riphah International University. Six surface EMG channels were recorded from each able-bodied subject. In case of amputees, one subject had only five surface electrodes. Data was recorded with sampling frequency of 8000 Hz and was filtered with analogue bandpass filter of bandwidth 10-500 Hz.

Each subject performed eleven movements for two consecutive days and on individual day, each movement was repeated four times with a contraction and relaxation time of five secs each. Movements included hand open, hand close, flex hand, extend hand, pronation, supination, side grip, fine grip, agree, pointer and rest. Each session took 400 secs to complete and sequence of movements was randomized for each session.

B. Data Processing

EMG signals are mostly contaminated by different sources of noise and it affects the performance of classifier. Therefore, after initial digital bandpass filtering with Butterworth 3rd order filter of bandwidth 20-500 Hz, power spectral density of the data was analyzed as shown in figure 1. It was found that signal included a noise of 50 Hz and its harmonics that were randomly distributed for each subject. Therefore, a 3rd order configurable Butterworth bandstop filter was specifically used to eliminate and reduce this noise.

After filtering, windows of 200 msec were chosen with window gap of 28.5 msec and four features including,

waveform length (WL), zero crossing (ZC), slope sign change (SSC) and moving average value (MAV) were extracted from each window. For classification, both the autoencoders and LDA classifiers are used. Autoencoders used in this study contained two layers each with length of 24 and 12 units respectively. Parameters were tuned for both layers and were stacked with final softmax classifier layer. The network was fine-tuned before classification and its performance is compared with the state of art LDA classifier which was developed with the purpose to provide benchmark for other similar algorithms. Two types of analysis including within day and between days' performance of classifiers were evaluated. For within day analysis, five-fold cross validation scheme was used while for between days' analysis, four-fold scheme was used. All these steps were applied in exactly the same way for both healthy and amputee subjects.

In order to evaluate the performance difference of two classifiers, Friedman's statistical test was used and p values less than 0.05 were considered significant.

III. RESULTS

A. Within day Analysis

In this analysis, for an individual subject, data of a single day was randomly divided into five equal folds. Four folds were used for training the classifier and one-fold for testing. This was repeated five times and each time, a new fold was used for testing. Classification errors from each repetition were averaged for a single day and for an individual subject, classification error was calculated as mean of two days. Results are presented as mean of all five healthy and two amputee subjects separately and are shown in figure 2.

To compare the performance of both kinds of data, statistical tests were conducted and P-values are tabulated in table1.

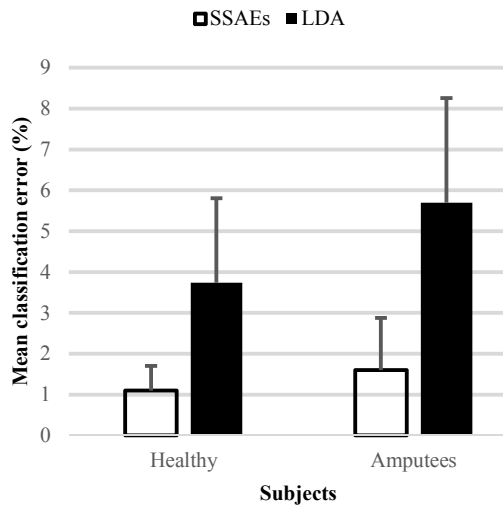


Fig. 2. Mean classification errors of within day analysis for both the healthy and amputee subjects obtained with both SSAEs and LDA. Bars represent the standard deviation between ten healthy and six amputee subjects' data.

SSAEs achieved comparatively less error rates than LDA and outperformed it for both healthy and amputee subjects. As can be seen from table 1, there was a less error difference between the healthy and amputee subject's data using SSAEs while this difference was quite enough for LDA as compared to SSAEs with percentage points of 2.02. Hence, SSAEs significantly improved the performance for amputee subjects as compared to LDA.

B. Between days Analysis

In this analysis, data of two days was randomly divided into four equal folds. Three folds were used for training the classifier and one-fold was used for testing. It was repeated four times and each time a new fold was used for testing. Classification error for an individual subject are calculated as mean of four repetitions and overall results are presented as

TABLE 1. FRIEDMAN TEST FOR WITHIN DAY ANALYSIS. P VALUES WERE CALCULATED TO QUANTIFY THE DIFFERENCE OF CLASSIFIERS FOR BOTH HEALTHY AND AMPUTEE SUBJECTS

Subjects	Mean accuracy $\pm$ standard deviation		P value
	SSAEs	LDA	
healthy	98.90 $\pm$ 0.60	96.26 $\pm$ 2.07	0.0017
amputee	98.40 $\pm$ 1.28	94.27 $\pm$ 2.56	0.0020

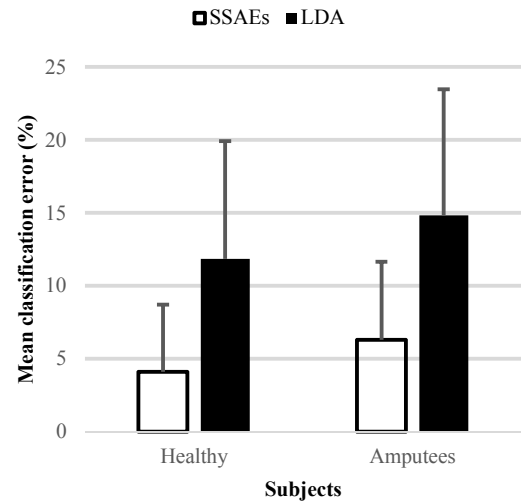


Fig. 3. Mean classification errors of Between days' analysis for both the healthy and amputee subjects obtained with both SSAEs and LDA. Bars represent the standard deviation between ten healthy and six amputee subjects' data.

mean of all five healthy and two amputee subjects and are shown in figure 3.

As can be seen from table 2, SSAEs outperformed the LDA in between days' analysis for both the healthy and amputee subjects. There was a less variation between the error rates of both healthy and amputee subjects while this variation was almost double with LDA classifier as compared to SSAEs.

IV. DISCUSSION

In recent years, deep learning techniques showed promising results in various fields like computer vision and natural language processing [29]. Various architectures of deep learning have also been applied to biomedical signals including EEG, ECG and medical image processing. The purpose of this study was to investigate the performance of SSAEs, an emerging deep learning technique, for classification of different hand movements. Therefore, sEMG data of five able and two amputee subjects was collected for consecutive 2 days and performance of SSAEs was compared with conventional machine learning algorithm for two different types of analysis.

In the within day analysis, SSAEs outperformed the LDA with p values 0.0017 and 0.0020 for both able-bodied and amputee subjects respectively. Autoencoders improved the performance for amputee subjects and achieved close error rates to that of healthy subjects. While using LDA, the percentage difference between the healthy and amputees' data was 2.02.

For between days analysis, SSAEs again outperformed ($P < 0.001$) the LDA for both the data of healthy and amputee subjects. The percentage points difference between the classification errors of both healthy and amputee subjects were

TABLE 2. FRIEDMAN TEST FOR BETWEEN DAYS' ANALYSIS. P VALUES WERE CALCULATED TO QUANTIFY THE DIFFERENCE OF CLASSIFIERS FOR BOTH HEALTHY AND AMPUTEE SUBJECTS.

Subjects	Mean accuracy $\pm$ standard deviation		P value
	SSAEs	LDA	
healthy	95.90 $\pm$ 4.60	88.16 $\pm$ 8.07	2.53e-10
amputee	93.70 $\pm$ 5.35	85.17 $\pm$ 8.65	9.63e-07

2.20 and 3.0 using SSAEs and LDA respectively. For healthy subjects, SSAEs achieved close error rates for both the within and between days analysis with percentage points difference of 3.0 while this difference for LDA was 8.10. Similarly, this difference for amputee subjects was 4.70 and 9.08 percentage points for both SSAEs and LDA respectively. Autoencoders significantly improved the performance for both healthy and amputee subjects and even achieved comparatively less error rates for between days analysis as compared to LDA and proved to be robust enough for long term assessment.

V. CONCLUSION

This study investigated the performance of SSAEs (an emerging deep learning architecture) against the conventional machine learning algorithm for classification of eleven hand movements. The results showed that SSAEs significantly improved the performance for both the healthy and amputee subjects and it proved to be more reliable than LDA for both types of analysis.

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